

PRASAD LOKDU JADHAV

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EDUCATION

Shri Guru Gobind Singhji Institute of Engineering and Technology

B.Tech in Information Technology — CGPA: 9.0/10.0

Nanded, India

Aug 2023 – May 2027

EXPERIENCE

Indian Institute of Technology Hyderabad (IITH)

5G Uplink RAN Slicing Research Intern

Hyderabad, India

May 2026 – Present

- Configuring and deploying simulated **5G NR uplink environments** using **OpenAirInterface (OAI)** in RF simulator mode for uplink scheduling experiments.
- Developing Python-based MAC log parsers extracting per-TTI scheduling metrics including **RBG allocation, BSR reports, and uplink throughput statistics**.
- Modifying **OAI uplink scheduler (nr_ulsch_preprocessor)** to enforce fixed resource block allocation policies for controlled experimentation.
- Automating bursty multi-UE uplink traffic generation using **iperf3, Bash scripting, subprocess automation, and Linux networking tools**.
- Analyzing scheduling fairness, throughput efficiency, and BSR inaccuracies under heterogeneous channel conditions using **Matplotlib, statistical analysis, and visualization pipelines**.

PROJECTS

PulseNet – Distributed Real-Time Chat System | *Go, NestJS, TypeScript, PostgreSQL, Redis, Pub/Sub, WebSockets, Next.js*

- Architected a horizontally scalable **real-time distributed chat platform** using **event-driven microservices**, Redis Pub/Sub, and WebSocket-based bidirectional communication handling concurrent messaging workloads.
- Designed **distributed message delivery pipelines**, session management, and PostgreSQL persistence layers with Redis caching to optimize **low-latency real-time communication**.
- Implemented **JWT authentication**, refresh token rotation, API gateway communication, and Redis-based **rate limiting** using NestJS and Golang services.
- Built scalable **asynchronous event-processing workflows** improving hot-path message delivery efficiency and reliability across distributed backend services.

High Priority Distributed Job Queue | *Go, PostgreSQL, Docker, Goroutines, Shell Scripting*

- Engineered a **concurrent distributed job queue** with worker-pool architecture, **priority scheduling**, retry handling, and transactional job-state management.
- Implemented PostgreSQL-backed **durable task orchestration** supporting pending, inflight, retry, and completed state transitions with **crash-safe recovery mechanisms**.
- Processed **1000+ jobs at 19 jobs/sec** during load testing while achieving **93% successful execution** with bounded retries and visibility-timeout reprocessing.
- Developed **operational monitoring endpoints** exposing queue health metrics, throughput statistics, and worker execution observability for distributed systems monitoring.

RAG SaaS Applications | *Python, RAG Pipelines, Vector Search, LLMs, FastAPI, Prompt Engineering*

- Built **retrieval-augmented generation (RAG)** applications integrating semantic vector search, document chunking, embedding pipelines, and **LLM orchestration workflows**.
- Designed AI-powered **contextual querying systems** supporting document understanding, grounded response generation, and **multi-stage retrieval pipelines**.
- Implemented modular **prompt-engineering workflows** and retrieval evaluation pipelines improving response grounding, hallucination reduction, and retrieval accuracy.
- Developed scalable **FastAPI backend services** for ingestion, retrieval, and context-aware generation using modular AI system architecture.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, JavaScript, C++, SQL

Backend & Systems: Distributed Systems, Microservices, REST APIs, WebSockets, Event-Driven Architecture, Concurrent Systems, Goroutines, JWT Auth, Rate Limiting

AI / ML: Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM APIs, Vector Search, AI Workflows

Databases & Infrastructure: PostgreSQL, Redis, MongoDB, Docker, Linux, Pub/Sub, GitHub Actions

Networking & Research: OpenAirInterface (OAI), 5G NR, MAC Scheduling, RF Simulation, Wireless Networking, iperf3

Tools & Libraries: NestJS, Next.js, React, Matplotlib, Pandas, NumPy, Bash, Regex, Git

ACHIEVEMENTS

- Solved **450+ DSA problems** across LeetCode and competitive programming platforms with **1700+ LeetCode rating**.
- Achieved **1000+ Codeforces rating** and secured **Top 10 position in HackSpectra Hackathon**.
- Completed **Full Stack Open Certification** covering React, TypeScript, Node.js, GraphQL, testing, and CI/CD workflows.
- Completed **5G Security ISEA Bootcamp** focused on wireless security, 5G infrastructure, and telecom network concepts.